Test Series No.

01

All India Test Series
NEET (2025-26)

DURATION: 180 Minutes | **DROPPER** | **M. MARKS: 720**

DATE: 14/09/2025

ANSWER KEY

1. (3)	46. (4)	91. (2)	136. (2)
2. (3)	47. (2)	92. (3)	137. (4)
3. (3)	48. (2)	93. (3)	138. (2)
4. (4)	49. (1)	94. (3)	139. (3)
5. (3)	50. (4)	95. (1)	140. (1)
6. (4)	51. (3)	96. (2)	141. (2)
7. (4)	52. (1)	97. (2)	142. (3)
8. (1)	53. (3)	98. (2)	143. (2)
9. (4)	54. (2)	99. (4)	144. (3)
10. (4)	55. (2)	100. (2)	145. (4)
11. (1)	56. (4)	101. (2)	146. (4)
12. (2)	57. (2)	102. (2)	147. (2)
13. (3)	58. (1)	103. (2)	148. (4)
14. (4)	59. (1)	104. (4)	149. (3)
15. (2)	60. (3)	105. (4)	150. (1)
16. (3)	61. (1)	106. (3)	151. (4)
17. (4)	62. (3)	107. (1)	152. (1)
18. (2)	63. (2)	108. (1)	153. (3)
19. (4)	64. (3)	109. (3)	154. (2)
20. (4)	65. (2)	110. (2)	155. (2)
21. (2)	66. (2)	111. (3)	156. (4)
22. (3)	67. (1)	112. (2)	157. (4)
23. (3)	68. (2)	113. (2)	158. (4)
24. (4)	69. (4)	114. (2)	159. (1)
25. (3)	70. (1)	115. (3)	160. (4)
26. (3)	71. (2)	116. (1)	161. (4)
27. (2)	72. (2)	117. (4)	162. (3)
28. (3)	73. (4)	118. (2)	163. (4)
29. (2)	74. (4)	119. (4)	164. (2)
30. (2)	75. (1)	120. (2)	165. (3)
31. (3)	76. (1)	121. (3)	166. (3)
32. (4)	77. (1)	122. (3)	167. (3)
33. (4)	78. (1)	123. (2)	168. (3)
34. (4)	79. (3)	124. (1)	169. (3)
35. (3)	80. (2)	125. (2)	170. (3)
36. (2)	81. (1)	126. (3)	171. (3)
37. (2)	82. (4)	127. (4)	172. (3)
38. (2)	83. (2)	128. (3)	173. (4)
39. (2)	84. (4)	129. (2)	174. (4)
40. (3)	85. (4)	130. (4)	175. (4)
41. (2)	86. (3)	131. (2)	176. (3)
42. (4)	87. (4)	132. (2)	177. (2)
43. (2)	88. (2)	133. (3)	178. (1)
44. (2)	89. (2)	134. (3)	179. (3)
45. (4)	90. (3)	135. (2)	180. (1)

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1. (3)

Here one division along the horizontal is l.

Projection along the inclined is $l \sec\theta$,

so least count = 1 MSD – 1VSD

$= l \sec\theta - l$

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2. (3)

v = 4 – 2x

$$\frac{dx}{dt} = 4 - 2x$$

$$\Rightarrow \int_0^x \frac{dx}{4-2x} = \int_0^t dt$$

$$-\frac{1}{2}\left|\ln(4-2x)\right|_0^x = t$$

$$\Rightarrow \ln\left(\frac{4-2x}{4}\right) = -2t$$

$$\Rightarrow x = 2(1-e^{-2t})$$

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3. (3)

Along AB: $s = 2R,\ a = g \Rightarrow t = \sqrt{\frac{4R}{g}}$

Along AC: angle = 60° $\Rightarrow a = g \sin 60°$;

chord $AC = 2R \sin 60° = \sqrt{3}R$

Hence $t = \sqrt{\frac{2s}{a}} = \sqrt{\frac{4R}{g}}$.

So (1) and (2) are true.

Speeds: $v_B = \sqrt{4gR}$,

$v_C = \sqrt{2(g\sin 60°)(2R\sin 60°)} = \sqrt{3gR} \neq v_B$

Thus (3) is the incorrect statement.

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4. (4)

Angle swept from A to B (clockwise):

$$\theta = 30° + 60° + 120° = 210° = \frac{7\pi}{6}\text{rad}$$

$$\vec{v}_A = \frac{v}{2}\hat{i} - \frac{\sqrt{3}v}{2}\hat{j}$$

$$\vec{v}_B = \frac{-\sqrt{3}v}{2}\hat{i} + \frac{v}{2}\hat{j}$$

$$\Delta\vec{v} = \vec{v}_B - \vec{v}_A$$

$$= -\frac{v}{2}\left(\sqrt{3}+1\right)\hat{i} + \frac{v}{2}\left(\sqrt{3}+1\right)\hat{j}$$

$$|\Delta\vec{v}| = \frac{v}{\sqrt{2}}\left(\sqrt{3}+1\right) = \frac{v}{2}\left(\sqrt{6}+\sqrt{2}\right)$$

$$\Delta t = \frac{R\theta}{v} = \frac{7\pi R}{6v}$$

Average acceleration

$$|\vec{a}_{av}| = \frac{|\Delta\vec{v}|}{\Delta t} = \frac{\frac{v\left(\sqrt{6}+\sqrt{2}\right)}{2}}{\frac{7\pi R}{6v}}$$

$$\frac{6v^2\left(\sqrt{6}+\sqrt{2}\right)}{14\pi R} = \frac{6\left(\sqrt{3}+1\right)v^2}{7\sqrt{2}\pi R}$$

$$|\vec{a}_{av}| = \frac{6\left(\sqrt{3}+1\right)v^2}{7\sqrt{2}\pi R}$$

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5. (3)

$$v_x = \frac{dx}{dt} = 4 \text{ (constant).}$$

$$v_y = \frac{dy}{dt} = 2.\frac{dx}{dt} - 2x.\frac{dx}{dt} = 2v_x(1-x)$$

$$a_y = \frac{dv_y}{dt} = 2v_x(-v_x) = -2v_x^2 = -2(4)^2 = -32 \text{ m s}^{-2}$$

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6. (4)

Time for 20 = 40 ± 1 s $\Rightarrow T = \frac{40}{20} = 2.0$ s

$$\frac{\Delta T}{T} = \frac{\Delta t}{t} \Rightarrow \frac{\Delta T}{T} \times 100 = \frac{1}{40} \times 100 = 2.5\%$$

For a pendulum, $g = 4\pi^2 \frac{l}{T^2}$

$\Rightarrow$ fractional error $\frac{\Delta g}{g} = \frac{\Delta l}{l} + 2\frac{\Delta T}{T}$.

$\Delta l/l = 0.1/100 = 0.001 = 0.1\%$

$\Rightarrow$ % error in g = 0.1% + 5.0% = 5.1%

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7. (4)

At projection, speed = u.

At the highest point, $v_x = u\cos\theta_0$ and $v_y = 0$

$\Rightarrow$ speed = $u\cos\theta_0$.

magnitude of change in magnitude of velocity (speed)

$= u - u\cos\theta_0 = u(1 - \cos\theta_0)$.

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[Contd...

8. (1)

$$P=\frac{a^2b}{\sqrt{cd^3}}$$

$$\Rightarrow \frac{\Delta P}{P}=2\left(\frac{\Delta a}{a}\right)+\left(\frac{\Delta b}{b}\right)+\frac{1}{2}\left(\frac{\Delta c}{c}\right)+\frac{3}{2}\left(\frac{\Delta d}{d}\right)$$

$$\Rightarrow \frac{\Delta P}{P}\times 100 = 2\times 2\% + 1\% + \frac{1}{2}\times 1\% + \frac{3}{2}\times 2\%$$

$= 4\% + 1\% + 0.5\% + 3\% = 8.5\%$

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9. (4)

$\cos\left(60^\circ\right)=\frac{1}{2}$ is a constant

$$\Rightarrow \frac{d}{dx}\left[\cos\left(60^\circ\right)\right]=\frac{d}{dx}\left[\frac{1}{2}\right]=0$$

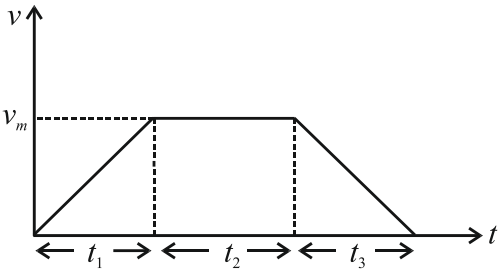
10. (4)

(A) is false: in projectile motion the magnitude of $\frac{dv}{dt}$ (acceleration) is constant and equals g.

(R) is true: the speed decreases up to the highest point and then increases due to gravity.

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11. (1)



Let maximum speed be v.

$t_1 = v_m/2$, $t_3 = v_m/2$ (since $a = 2$, retardation = 2);

time $t_2 = 10 - v_m$.

Distance S = area under $v - t$ graph

$$S=\frac{v_m^2}{2}+(10-v_m)\times v_m = 10v_m - \frac{{v_m}^2}{2} = 32$$

$$\Rightarrow {v_m}^2 - 20v_m + 64 = 0 \Rightarrow v_m = 4\ \text{m s}^{-1}$$

(discard $v_m = 16$ since $t_2 < 0$).

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12. (2)

Uniform circular motion ⇒ speed is constant → $\frac{d|\vec{v}|}{dt}=0$.

But the velocity direction changes, so

$$\left|\frac{d\vec{v}}{dt}\right| = a_c = \frac{v^2}{R} \neq 0$$

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13. (3)

$$\text{Least count} = \frac{\text{pitch}}{\text{No. of CSD}}$$

Pitch = 100 × 0.01 = 1 mm = 0.1 cm

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14. (4)

For same range, angles of projections are θ and (90 – θ) for same speed of projection.

$$H_1=\frac{u^2\sin^2\theta}{2g},$$

$$H_2=\frac{u^2\sin^2(90-\theta)}{2g}=\frac{u^2\cos^2\theta}{2g}$$

$$H_1+H_2=\frac{u^2\sin^2\theta}{2g}+\frac{u^2\cos^2\theta}{2g}$$

$$= \frac{u^2}{2g}=\frac{(20)^2}{2\times 10}=20\ \text{m}$$

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15. (2)

$$K=\frac{Ar^{2/3}}{B+t^3}\Rightarrow [B]=T^3 \text{ and}$$

$$[A]=[K][B]L^{-2/3}=\left(ML^2T^{-2}\right)\left(T^3\right)L^{-2/3}=[ML^{4/3}T]$$

Let $F = A^xB^yL^z$ have dimensions of force MLT^{-2}.

Matching powers (dimensional analysis):

$$M: x = 1; L: \frac{4}{3}x+z=1\Rightarrow z=-\frac{1}{3};$$

$$T: x+3y=-2\Rightarrow y=-1$$

Therefore $x^2-y+z=1-(-1)-\frac{1}{3}=\frac{5}{3}$

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16. (3)

$$u_x = u\cos 53^\circ = 100\,\text{m s}^{-1},$$

$$u_y = u\sin 53^\circ = \frac{400}{3}\,\text{m s}^{-1}\ \text{downward.}$$

Vertical motion:

$$H=\frac{5500}{3}=u_yt+\frac{1}{2}gt^2\Rightarrow 15t^2+400t-5500=0$$

$$\Rightarrow t = 10\text{s}$$

Horizontal range = $= u_xt = 100\times 10 = 1000$ m

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[Contd...

17. (4)

$x = 3t^2 - \frac{t^3}{2} \Rightarrow v = \dot{x} = 6t - \frac{3}{2}t^2 = 1.5t(4-t)$,

$a = \frac{dv}{dt} = 6 - 3t$

v and a are in the same direction if $va > 0$;

$\Rightarrow 0 < t < 2 \rightarrow$ Q.

Particle is at origin $\Rightarrow x = 0 \Rightarrow t = 0, 6$ s $\rightarrow$ R.

Retarding if $\Rightarrow va < 0 \Rightarrow 2$ s $< t < 4$ s $\rightarrow$ S.

Speed zero $v = 0 \Rightarrow t = 0, 4$ s $\rightarrow$ P.

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18. (2)

For constant acceleration: $v^2 = u^2 + 2ax$.

Here $u^2 = 20,\ a = -2 \Rightarrow v = \sqrt{20 - 4x}$, which decreases as a square-root and becomes zero at x = 5 m.

So v vs x is a concave-down curve hitting the axis at 5 m $\rightarrow$ graph (2).

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19. (4)

Upward motion: net downward a = g + 2 = 12

$\Rightarrow t_a = \frac{u}{12} = \frac{40}{12} = \frac{10}{3}$ s and

$H = \frac{u^2}{2(12)} = \frac{200}{3}$ m

$\Rightarrow$ (1) & (3) false.

Downward motion: net downward a = g – 2 = 8

$t_d = \sqrt{\frac{2H}{a}} = \sqrt{\frac{2(200/3)}{8}} = 5\sqrt{\frac{2}{3}}$ s $\Rightarrow$ (2) is false.

Speed just before hitting ground :

$v = \sqrt{2aH} = 40\sqrt{\frac{2}{3}}$ m s^{-1} $\Rightarrow$ (4) true.

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20. (4)

$\vec{A}$ and $\vec{B}$ are perpendicular $\Rightarrow \vec{A} \cdot \vec{B} = 0$:

$(\alpha\hat{i} - 2\hat{j} + 4\hat{k}) \cdot (\alpha\hat{i} - \alpha\hat{j} - 2\hat{k}) = \alpha^2 + 2\alpha - 8 = 0$

$\Rightarrow (\alpha - 2)(\alpha + 4) = 0$

So $\alpha = 2$ or $\alpha = -4$.

21. (2)

For a parallelogram with diagonals D_1, D_2:

$\text{Area} = \frac{1}{2}|\vec{D}_1 \times \vec{D}_2|$

$\vec{D}_1 = (6\hat{i} + 4\hat{j}),\ \vec{D}_2 = (-4\hat{j}) \Rightarrow \vec{D}_1 \times \vec{D}_2 = -24\hat{k}$

So area $= \frac{1}{2} \times 24 = 12$ m^2

22. (3)

At top: $v_{\max H} = u\cos\theta$

At $y = \frac{H}{2}$:

$v_y^2 = u^2\sin^2\theta - 2g\left(\frac{u^2\sin^2\theta}{4g}\right) = \frac{u^2\sin^2\theta}{2}$

$v_{\text{net}}^2 = u^2\cos^2\theta + \left(u\sin\theta/\sqrt{2}\right)^2$

$= u^2\left(\cos^2\theta + \frac{1}{2}\sin^2\theta\right)$

Given $u\cos\theta = \sqrt{\frac{6}{7}}u\sqrt{\cos^2\theta + \frac{1}{2}\sin^2\theta}$

$\Rightarrow \cos^2\theta = \frac{6}{7}\left(\cos^2\theta + \frac{\sin^2\theta}{2}\right)$

$\Rightarrow \cos^2\theta = 3\sin^2\theta$

$\Rightarrow \tan^2\theta = \frac{1}{3}$

So $\theta = 30°$.

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23. (3)

When particle is moving with constant velocity, average velocity is equal to instantaneous velocity.

$\vec{a} = \frac{d\vec{v}}{dt} = \vec{0}$

As speed is equal to magnitude of velocity, distance covered is equal to magnitude of displacement in any time interval.

So, (A), (B), (C) and (D) all are correct.

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24. (4)

From engine at pole to tail at pole:

displacement

$= L$, speeds $u = 5$, $v = 10$.

Using $v^2 - u^2 = 2aL \Rightarrow aL = \frac{100 - 25}{2} = 37.5$

When 2/3 of the train is behind the pole, only L/3 has passed $\Rightarrow$ s = L/3.

Then $v^2 = u^2 + 2a(L/3) = 25 + \frac{75}{3} = 50$

$\Rightarrow v = 5\sqrt{2}$ m s^{-1}

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25. (3)

Sum: 436.32 + 227.2 + 0.301 = 663.821 → keep 1 decimal (least among addends) ⇒ 663.8

$v = \frac{12.0}{3.0} = 4.0\ \text{m s}^{-1}$. Fractional error.

$\frac{\Delta v}{v} = \frac{0.1}{12.0} + \frac{0.1}{3.0} \approx 0.0417 \Rightarrow \Delta v \approx 0.17 \approx 0.2$

So $v = (4.0 \pm 0.2)\ \text{m s}^{-1} \neq (3.00 \pm 0.10)$

Statement I true, II false.

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26. (3)

Addition/subtraction requires same dimensions. Since *A, B, C* are pairwise different, (1), (2) and (4) are invalid. But *AB* + *C* can be meaningful if [*AB*] = [*C*]

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27. (2)

Using dimensional relation

$F = MLT^{-2} \Rightarrow M = \frac{FT^2}{L}$

Unit mass in new system:

$M' = \frac{(100\,\text{N})(60\,\text{s})^2}{10\,\text{m}} = 36{,}000\,\text{kg}$.

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28. (3)

$y = x^3 - 12x \Rightarrow \frac{dy}{dx} = 3x^2 - 12 = 0$ at $x = \pm 2$.

$\frac{d^2y}{dx^2} = 6x \Rightarrow \frac{d^2y}{dx^2} = 12 > 0 \Rightarrow y$ is minimum at $x = 2$.

$y_{min} = 2^3 - 12 \times 2 = -16$

29. (2)

Rain appears vertical when man moves at 3 m s^{-1}

⇒ horizontal component of rain $v_x = 3$.

With man at 6 m s^{-1}, apparent path makes 30° with vertical:

$\tan 30^\circ = \frac{|v_x - 6|}{v_y} = \frac{3}{v_y} \Rightarrow v_y = 3\sqrt{3}$

Actual speed

$v = \sqrt{v_x^2 + v_y^2} = \sqrt{3^2 + (3\sqrt{3})^2} = 6\,\text{m s}^{-1}$

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30. (2)

$\int_0^{\pi/4} (\cos^2 x - \sin^2 x)\,dx = \int_0^{\pi/4} \cos 2x\,dx$

$= \left[\frac{1}{2}\sin 2x\right]_0^{\pi/4}$

$= \frac{1}{2}(1 - 0) = \frac{1}{2}$

31. (3)

$y = 4x - 2x^2 \Rightarrow \frac{dy}{dx} = 4 - 4x.$

At $x = 0$: slope = 4; at $x = 1$: slope = 0.

32. (4)

Take +*x* along stream, +*y* across.

Swimmer's velocity (relative to water) at 127° to downstream:

$u_x = 5\cos 127^\circ = -3$ m/s, $u_y = 5\sin 127^\circ = 4$ m/s.

Net x-velocity = –3 + 4 = 1 m/s;

crossing time t = 500/4 = 125 s

⇒ drift = 1 × 125 = 125 m.

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33. (4)

$\vec{R} = \vec{A} + \vec{B}$ and

$\vec{R} \perp \vec{A} \Rightarrow (\vec{A} + \vec{B}) \cdot \vec{A} = 0 \Rightarrow \vec{A} \cdot \vec{B} = -A^2.$

$|\vec{R}|^2 = |\vec{A} + \vec{B}|^2 = A^2 + B^2 + 2\vec{A} \cdot \vec{B} = B^2 - A^2 = 36.$

With $A + B = 18 \Rightarrow B - A = 2 \rightarrow A = 8, B = 10.$

$\cos\theta = \frac{\vec{A} \cdot \vec{B}}{AB} = -\frac{A}{B} = -\frac{8}{10} = -\frac{4}{5} \Rightarrow \theta \approx 143^\circ.$

34. (4)

Time to hit ground: solve $0 = 25 + 20t - \frac{1}{2}gt^2$

$\Rightarrow 0 = 25 + 20t - 5t^2 \Rightarrow t = 5\,\text{s}$

Distance travelled = 20 m up + 45 m down = 65 m

⇒ average speed = 65/5 = 13 m s^{-1}

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35. (3)

Relative motion (B w.r.t. A): $r_0 = 5$ m,

$u_{rel} = 10 - 20 = -10\ \text{m s}^{-1}$, $a_{rel} = 4 + a$

(if A brakes with retardation *a*).

For no colision,

$v_{rel} = 0$

$v_{rel}^2 = u_{rel}^2 + 2a_{rel}s_{rel}$

$a_{rel} = \frac{(10)^2}{2 \times 5} = 10$

⇒ 4 + a = 10

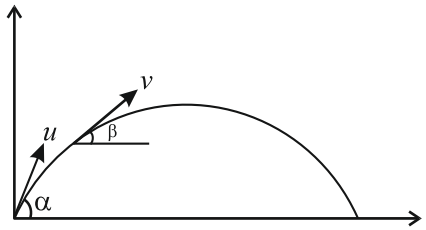
⇒ a = 6 m s^{-2}

So A must brake with at least 6 m s^{-2}.

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36. (2)



Let v be the speed of particle, when velocity vector makes angle β with the horizontal.

As $v_x = u_x$

$\Rightarrow v\cos\beta = u\cos\alpha$

$\Rightarrow v = u\cos\alpha\sec\beta$

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37. (2)

Since $a = \frac{dv}{dt} \Rightarrow dt = \frac{dv}{a}$, time $t = \int_2^4 \frac{dv}{a}$ = area under the $1/a$ –vs – v graph.

Between $v = 2$ and 4, $1/a$ rises linearly from 2 to 4 (units s^2/m).

Area (trapezium) $= \frac{(2+4)}{2} \times (4-2) = 6$ s.

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38. (2)

For collision, both particles should have equal vertical displacement in same time interval.

i.e. $s_{1y} = s_{2y}$

$\Rightarrow v_{1y}t - \frac{1}{2}gt^2 = v_2 t - \frac{1}{2}gt^2$

$\Rightarrow v_{1y} = v_2$

$\Rightarrow v_1 \sin 30° = v_2$

$\Rightarrow \frac{v_1}{v_2} = 2$

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39. (2)

$\frac{x}{3} + \frac{y}{4} = 1$

$\Rightarrow \frac{4x+3y}{12} = 1$

$\Rightarrow 3y = -4x + 12$

$\Rightarrow y = -\left(\frac{4}{3}x\right) + 4$

Which represents equation of straight line with slope with slopes $-\frac{4}{3}$ and intercept 4.

40. (3)

For constant acceleration with start from rest:

$v^2 = 2ax$ (from $v^2 - u^2 = 2ax$ with $u = 0$).

Hence $\frac{v^2}{x} = 2a$ = constant $\Rightarrow$ a horizontal line vs t.

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41. (2)

At long time (terminal speed), a = 0. From

$a = 8 - 2v \Rightarrow 0 = 8 - 2v$, so $v = 4$ m s^{-1}

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42. (4)

Let the boy start at $x = 0$: $x_{boy} = ut$.

Bus from rest with

a = 2 : $x_{bus} = 25 + \frac{1}{2}(2)t^2 = 25 + t^2$

To catch the bus, $x_{boy} = x_{bus}$

$\Rightarrow t^2 - ut + 25 = 0$.

For real t : $(-u)^2 - 4 \times 1 \times 25 \geq 0$

$u^2 \geq 100 \Rightarrow u_{min} = 10$ m s^{-1}

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43. (2)

Significant figures = all reliable (certain) digits plus the first uncertain (doubtful) digit.

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44. (2)

From the a–t graph:

$a = 10$ for $0 \leq t \leq 2 \Rightarrow \Delta v_1 = 10 \times 2 = 20$.

For $t \geq 2$, a falls linearly: $a(t) = 15 - 2.5t$.

With $v(0) = 0$:

$0 = \int_0^T a\,dt = 20 + \int_2^T (15 - 2.5t)\,dt$

$= 20 + 15T - 1.25T^2 - 25$

So $T^2 - 12T + 4 = 0 \Rightarrow T = 6 + 4\sqrt{2} \approx 11.65$ s

(take $T > 2$).

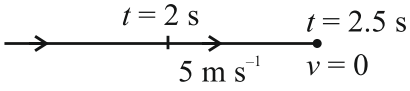
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45. (4)

$x = 25t - 5t^2$

$v = = 25 - 10t$

$v = 0$ at $t = 2.5$ s



Distance covered in the third second

$D = \left[\frac{1}{2} \times (5) \times 0.5\right] \times 2 = 2.5$ m

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[Contd...

46. (4)

In NO_2^-, oxidation state of nitrogen is +3 which can be increased upto +5 means it can be oxidised so it can behave as reducing agent.

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47. (2)

C_6H_6 :

Mole of carbon = 6

Mole of hydrogen = 6

Mole percentage of carbon $= \frac{6}{6+6} \times 100 = 50\%$

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48. (2)

$2\,MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$

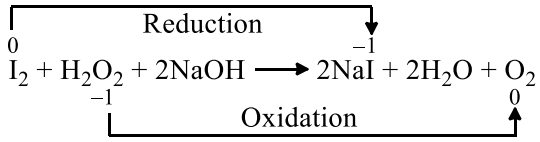
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49. (1)

Molarity and density depends on temperature as they involve volume term.

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50. (4)



Since different species are oxidised and reduced in the above reaction, so it is not a disproportionation reaction.

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51. (3)

- H_2: mole $= \frac{1}{2}$

  Volume at STP $= \frac{1}{2} \times 22.4 = 11.2$ L
- O_2: Mass = 0.2 × 32 = 6.4 g
- N_2: Mass = 1 × 28 = 28 g
- He: Atoms $= \frac{52}{4} = 13$

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52. (1)

- $Fe_3O_4 : FeO + Fe_2O_3$
- $Mn_3O_4 : 2MnO + MnO_2$
- $Pb_3O_4 : 2PbO + PbO_2$
- $C_3O_2 : O{=}C{=}C{=}C{=}O$

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53. (3)

Dalton's theory could not explain the laws of gaseous volumes.

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54. (2)

Layer test is used for the identification of Br^- and I^- ions.

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55. (2)

$$\underset{\substack{1\text{ mol}\\16\text{ g}}}{CH_4(g)} + \underset{\substack{2\text{ mol}\\2\times22.4\text{ L at STP}}}{2O_2(g)} \rightarrow CO_2(g) + 2H_2O(\ell)$$

Volume of O_2 required $= 2 \times 22.4 \times \frac{8}{16} = 22.4$ L

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56. (4)

- All decomposition reactions are not redox reactions.

  $\overset{+2\,+4\,-2}{CaCO_3} \rightarrow \overset{+2\,-2}{CaO} + \overset{+4\,-2}{CO_2}$
- Iodine gives an intense blue colour with starch.

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57. (2)

Molality $(m) = \frac{\left(\frac{10}{60}\right)}{90 \times 10^{-3}} = 1.85$ m

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58. (1)

$Ca(H_2PO_2)_2 : +2 + 2[2(+1) + x + 2(-2)] = 0$

$x = +1$

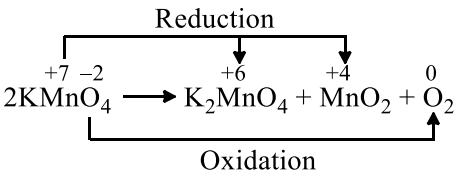
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59. (1)

Law of multiple proportions is applicable only when different compounds contain same elements.

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60. (3)



- Since moles are increasing in the reaction, so it is an example of decomposition reaction.
- Since oxidation and reduction both take place simultaneously so it is a redox reaction.

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[Contd...

61. (1)

In number written in scientific notation, all digits are significant e.g. 2.01×10^2 has three significant figures.

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62. (3)

- $NH_3 : x + 3(+1) = 0 \quad \therefore\ x = -3$
- $N_3H : 3x + 1 = 0 \quad \therefore\ x = -\frac{1}{3}$
- $N_2H_4 : 2x + 4(+1) = 0 \therefore\ x = -2$

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63. (2)

Element	Moles	Simple ratio
C	$\frac{25}{12} \simeq 2$	$\frac{2}{2} = 1$
H	$\frac{4}{1} = 4$	$\frac{4}{2} = 2$
Cl	$\frac{71}{35.5} = 2$	$\frac{2}{2} = 1$

$\therefore$ empirical formula $= CH_2Cl$

$$n = \frac{\text{Molar mass}}{\text{Empirical formula mass}} = \frac{99}{49.5} = 2$$

$\therefore$ Molecular formula = $(CH_2Cl)_n = C_2H_4Cl_2$

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64. (3)

Among halogens, F_2 is most powerful oxidising agent as reduction potential of fluorine is maximum.

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65. (2)

$$°F = \frac{9}{5}(°C) + 32 = \frac{9}{5} \times 0 + 32 = 32$$

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66. (2)

- Peroxide: $O_2^{2-} : 2x = -2 \Rightarrow x = -1$
- Ozonide: $O_3^- : 3x = -1 \Rightarrow x = -\frac{1}{3}$
- Oxide: $O^{2-} : x = -2$
- Superoxide: $O_2^- : 2x = -1 \Rightarrow x = -\frac{1}{2}$

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67. (1)

$$\text{Mole of } N^{3-} = \frac{0.7}{14} = 0.05$$

Number of valence electrons

$= 0.05 \times (5+3) \times N_A = 0.4\, N_A$

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68. (2)

- By convention, $E^o_{H^+/H_2} = 0.00$ volts
- The oxidation state of hydrogen is –1 when it is bonded to metals in binary compounds.

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69. (4)

$$\%\text{ of C} = \frac{12}{44} \times 100 = 27.27\%$$

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70. (1)

By comparison, Zn releases electrons to copper and copper releases electrons to silver.

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71. (2)

$$\text{Mass of one molecule} = \frac{\text{molar mass}}{N_A} = \frac{64}{N_A}\text{g}$$

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72. (2)

$HAuCl_4$ can be represented as $HAu(III)Cl_4$.

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73. (4)

Multiple		Prefix
10^{-2}	:	Centi
10^{-6}	:	Micro
10^{-12}	:	Pico
10^{-9}	:	Nano

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74. (4)

$$Cr_2O_7^{2-} \xrightarrow{\text{n-factor} = 6} 2Cr^{3+}$$

$$I^- \xrightarrow{\text{n-factor} = 1} \frac{1}{2} I_2$$

eq. of $K_2Cr_2O_7$ = eq. of I^-

$$n \times 6 = 1 \times 1 \Rightarrow n = \frac{1}{6}$$

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75. (1)

$n_i = n_{left} + n_{removed}$

$= 1.8 \times 10^{-3} + \frac{3.011 \times 10^{21}}{6.022 \times 10^{23}}$

$= 1.8 \times 10^{-3} + 5 \times 10^{-3} = 6.8 \times 10^{-3}$

Initial mass $= 6.8 \times 10^{-3} \times 2$

$= 13.6 \times 10^{-3}\,\text{g} = 13.6\,\text{mg}$

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[Contd...

76. (1)

- Combination reaction: $C + O_2 \rightarrow CO_2$
- Non-redox reaction:
  $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
- Metal displacement redox reaction:
  $CuSO_4 + Zn \rightarrow Cu + ZnSO_4$
- Non-metal displacement redox reaction:
  $2Na + 2H_2O \rightarrow 2NaOH + H_2$

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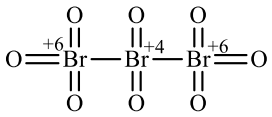
77. (1)

According to Dalton's atomic theory

- Matter consists of indivisible atoms.
- Atoms can neither be created nor destroyed in a chemical reaction.

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78. (1)



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79. (3)

$$M = \frac{\%\text{ by mass of solute} \times \text{density}\left(\frac{g}{mL}\right) \times 10}{\text{molar mass of solute}}$$

$$= \frac{36.5 \times 1.4 \times 10}{36.5} = 14\ M$$

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80. (2)

Carbon can exhibit oxidation states from –4 to +4.

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81. (1)

- H_2: No. of atoms $= \frac{4}{2} \times 2 \times N_A = 4\ N_A$
- O_3: No. of atoms $= \frac{32}{48} \times 3 \times N_A = 2\ N_A$
- Cl_2: No. of atoms $= \frac{71}{71} \times 2 \times N_A = 2\ N_A$
- Na: No. of atoms $= \frac{2.3}{23} \times 1 \times N_A = 0.1\ N_A$

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82. (4)

Oxidation can be defined as:

- addition of oxygen or electronegative element
- removal of hydrogen or electropositive element
- loss of electron(s)
- increase in oxidation number

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83. (2)

$$\underset{28\ g}{N_2} + 3H_2 \rightarrow \underset{6\ g}{2NH_3}$$

Hence, for 14 g N_2, required H_2 is 3 g
and for 7 g N_2, required H_2 is 1.5 g

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84. (4)

Fluorine does not show a disproportionation tendency because it can not show any positive oxidation state.

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85. (4)

Carbon exist as three isotopes ^{12}C, ^{13}C and ^{14}C with relative abundance 98.892%, 1.108% and 2×10^{-10} % respectively, whose average atomic mass is 12.011.

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86. (3)

$$\overset{+1\ -2}{2Cu_2O} + \overset{+1\ -2}{Cu_2S} \rightarrow \overset{0}{6Cu} + \overset{+4\ -2}{SO_2}$$

In this reaction, Cu_2O helps sulphur in Cu_2S to increase in its oxidation number, therefore, Cu(I) is an oxidant and sulphur of Cu_2S helps copper both in Cu_2S itself and Cu_2O to decrease its oxidation number, therefore, sulphur of Cu_2S is reductant.

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87. (4)

$$n = M \times V(L)$$

$$= 0.2 \times 100 \times 10^{-3} = 0.02$$

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88. (2)

$S_2F_2 : 2x + 2(-1) = 0$

$x = +1$

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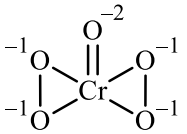
89. (2)

1 mol H_2O contain 16 g oxygen

$\therefore$ 0.2 mol H_2O will contain = 16 × 0.2
= 3.2 g oxygen.

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90. (3)



$x + 1(-2) + 4(-1) = 0 \Rightarrow x = +6$

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91. (2)

The content of nucleolus is continuous with the rest of the nucleoplasm as it is not a membrane bound structure.

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92. (3)

The cytoplasm is the main arena of cellular activities in both the plant and animal cells. Plant and animal cells are different as the former possess cell walls, plastids and a large central vacuole which are absent in animal cells. On the other hand, animal cells have centrioles which are absent in almost all plant cells.

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93. (3)

In some prokaryotes like cyanobacteria, there are membranous extensions into the cytoplasm called chromatophores which contain pigments. The interphase nucleus has a loose and indistinct network of nucleoprotein fibres called chromatin. Centromere holds two chromatids of a chromosome. In the chromoplasts, fat soluble carotenoid pigments like carotene, xanthophylls and others are present.

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94. (3)

The quasi-fluid nature of lipid enables lateral movement of proteins within the overall bilayer. This ability to move within the membrane is measured as its fluidity. The fluid nature of the membrane is also important from the point of view of functions like cell growth, formation of intercellular junctions, secretion, endocytosis, cell division etc.

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95. (1)

Typically, each chromosome has only one centromere. Every chromosome (visible only in dividing cells) essentially has a primary constriction or the centromere on the sides of which disc shaped structures called kinetochores are present.

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96. (2)

In 1838, Matthias Schleiden, a German botanist, examined a large number of plants and observed that all plants are composed of different kinds of cells which form the tissues of the plant. At about the same time, Theodore Schwann (1839), a German Zoologist, studied different types of animal cells and reported that cells had a thin outer layer which is today known as the 'plasma membrane'. He also concluded, based on his studies on plant tissues, that the presence of cell wall is a unique character of the plant cells.

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97. (2)

The core (axoneme) of a eukaryotic flagellum has a "9+2 array," which consists of nine doublets of peripheral microtubules and two central microtubules.

Centriole Structure: A centriole is made up of nine evenly spaced peripheral fibrils of tubulin protein. Each of the peripheral fibril is a triplet (meaning it's made of 3 microtubules). Therefore, the total number of microtubules is:

9 peripheral triplets × 3 microtubules per triplet = 27 microtubules.

The axoneme is the core of cilia and flagella, not centrioles. The central part of a centriole is called the "hub."

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98. (2)

Besides flagella, Pili and Fimbriae are also surface structures of the bacteria but do not play a role in motility.

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99. (4)

Ribosomes are non-membrane bound organelles found in all cells – both eukaryotic as well as prokaryotic. The eukaryotic ribosomes are 80S while the prokaryotic ribosomes are 70S. Each ribosome has two subunits, larger and smaller subunits. The two subunits of 80S ribosomes are 60S and 40S while that of 70S ribosomes are 50S and 30S. Here 'S' (Svedberg's Unit) stands for the sedimentation coefficient; it is indirectly a measure of density and size. Both 70S and 80S ribosomes are composed of two subunits. The ribosomes of the chloroplasts are smaller (70S) than the cytoplasmic ribosomes (80S). The mitochondrial matrix also possesses single circular DNA molecule, a few RNA molecules, ribosomes (70S) and the components required for the synthesis of proteins.

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100. (2)

The cell envelope consists of a tightly bound three layered structure i.e., the outermost glycocalyx followed by the cell wall and then the plasma membrane. Although each layer of the envelope performs distinct function, they act together as a single protective unit. The cell wall determines the shape of the cell and provides a strong structural support to prevent the bacterium from bursting or collapsing. Glycocalyx differs in composition and thickness among different bacteria. It could be a loose sheath called the slime layer in some, while in others it may be thick and tough, called the capsule.

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101. (2)

RER is frequently observed in the cells actively involved in protein synthesis and secretion. The smooth endoplasmic reticulum is the major site for synthesis of lipid.

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102. (2)

Every chromosome (visible only in dividing cells) essentially has a primary constriction or the centromere on the sides of which disc shaped structures called kinetochores are present. Sometimes a few chromosomes have non-staining secondary constrictions at a constant location. This gives the appearance of a small fragment called the satellite.

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103. (2)

In studying and understanding the physiological and behavioural processes, one can take a physico-chemical approach and use cell-free systems to investigate. This approach enables us to describe the various processes in molecular terms. The approach is established by analysis of living tissues for elements and compounds. It will tell us what types of organic compounds are present in living organisms. In the next stage, one can ask the question: What are these compounds doing inside a cell? And, in what way they carry out gross physiological processes like digestion, excretion, memory, defense, recognition, etc. In other words we answer the question, what is the molecular basis of all physiological processes? It can also explain the abnormal processes that occur during any diseased condition. This physico-chemical approach to study and understand living organisms is called 'Reductionist Biology'.

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104. (4)

Golgi apparatus is the important site of formation of glycoproteins and glycolipids. The isolated lysosomal vesicles have been found to be very rich in almost all types of hydrolytic enzymes. ER divides the intracellular space into two distinct compartments, i.e., luminal (inside ER) and extra luminal (cytoplasm) compartments. The vacuole is the membrane-bound space found in the cytoplasm. It contains water, sap, excretory product and other materials not useful for the cell.

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105. (4)

Typically a mitochondrion is sausage-shaped or cylindrical having a diameter of 0.2-1.0μm (average 0.5μm) and length 1.0-4.1μm.

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106. (3)

Centrosome is an organelle usually containing two cylindrical structures called centrioles. They are surrounded by amorphous pericentriolar materials. Both the centrioles in a centrosome lie perpendicular to each other in which each has an organisation like the cartwheel.

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107. (1)

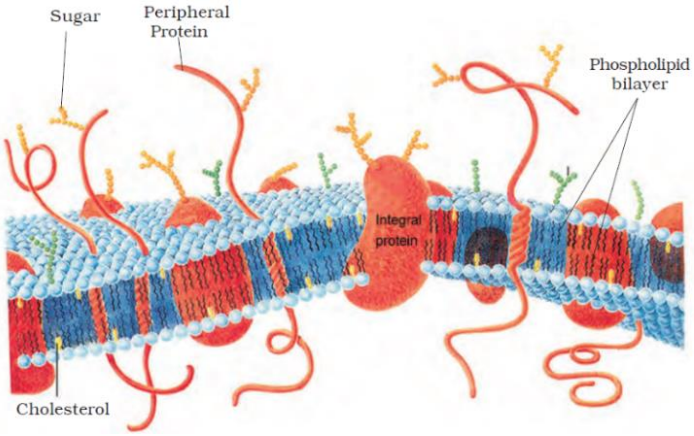
The chloroplasts are double membrane bound, with the inner membrane being less permeable. The ribosomes of the chloroplasts are 70S type. Chlorophyll pigments are located in the thylakoids. Chlorophyll pigments are present in the thylakoids.

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[Contd...

108. (1)

The lipids are arranged within the plasma membrane with the polar head (C) towards the outer sides and the hydrophobic tails towards the inner part. This ensures that the nonpolar tail of saturated hydrocarbons is protected from the aqueous environment. E is cholesterol. A is sugar, usually synthesised in chloroplasts, and in golgi bodies for glycosylation of proteins. Depending on the ease of extraction, membrane proteins can be classified as integral and peripheral. Peripheral proteins lie on the surface of membrane while the integral proteins are partially or totally buried in the membrane.



Fluid mosaic model of plasma membrane

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109. (3)

Schleiden and Schwann together formulated the cell theory. This theory however, did not explain as to how new cells were formed. Rudolf Virchow (1855) first explained that cells divided and new cells are formed from pre-existing cells (*Omnis cellula-e cellula*). Robert Brown discovered the nucleus. Camillo Golgi (1898) first observed densely stained reticular structures near the nucleus. These were later named Golgi bodies after him.

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110. (2)

Lysosomes: These are membrane bound vesicular structures formed by the process of packaging in the golgi apparatus. The isolated lysosomal vesicles have been found to be very rich in almost all types of hydrolytic enzymes (hydrolases – lipases, proteases, carbohydrases) optimally active at the acidic pH. These enzymes are capable of digesting carbohydrates, proteins, lipids and nucleic acids.

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111. (3)

In addition to the genomic DNA (the single chromosome/circular DNA), many bacteria have small circular DNA outside the genomic DNA. These smaller DNA are called plasmids. A special membranous structure is the mesosome which is formed by the extensions of plasma membrane into the cell. These extensions are in the form of vesicles, tubules and lamellae. They help in cell wall formation, DNA replication and distribution to daughter cells. They also help in respiration, secretion processes, to increase the surface area of the plasma membrane and enzymatic content.

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112. (2)

The endomembrane system include endoplasmic reticulum (ER), golgi complex, lysosomes and vacuoles. Since the functions of the mitochondria, chloroplast and peroxisomes are not coordinated with the above components, these are not considered as part of the endomembrane system.

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113. (2)

DNA is present in nucleus, mitochondria and chloroplasts. Many membrane bound minute vesicles called microbodies that contain various enzymes, are present in both plant and animal cells. The plasmid DNA confers certain unique phenotypic characters to such bacteria. One such character is resistance to antibiotics. Some mature cells even lack nucleus, e.g., erythrocytes of many mammals and sieve tube cells of vascular plants.

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114. (2)

Algae have cell wall, made of cellulose, galactans, mannans and minerals like calcium carbonate. They are eukaryotic organisms that are able to divide. All living cells possess cell membrane.

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[Contd...

115. (3)

As the polar molecules cannot pass through the nonpolar lipid bilayer, they require a carrier protein of the membrane to facilitate their transport across the membrane. A few ions or molecules are transported across the membrane against their concentration gradient, i.e., from lower to the higher concentration. Such a transport is an energy dependent process, in which ATP is utilised and is called active transport, e.g., Na^+/K^+ Pump.

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116. (1)

The eukaryotes include all the protists, plants, animals and fungi. Bacteria, cyanobacteria, mycoplasma all are prokaryotes.

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117. (4)

In *Amoeba,* the contractile vacuole is important for osmoregulation and excretion. ER helps in the transport of substances, synthesis of proteins, lipoproteins and glycogen.

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118. (2)

Mycoplasmas, the smallest cells, are only 0.3 µm in length. Nerve cells are some of the longest cells.

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119. (4)

A typical red blood cell is round and biconcave in shape. It lacks nucleus at maturity.

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120. (2)

The two mitochondrial membranes have their own specific enzymes associated with the mitochondrial function. Mitochondria are the sites of aerobic respiration. They produce cellular energy in the form of ATP, hence they are called 'power houses' of the cell.

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121. (3)

Unicellular organisms are capable of (i) independent existence and (ii) performing the essential functions of life. Anything less than a complete structure of a cell does not ensure independent living. Hence, cell is the fundamental structural and functional unit of all living organisms.

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122. (3)

The Golgi cisternae consist of many flat, disc-shaped sacs or cisternae of 0.5µm to 1.0µm diameter.

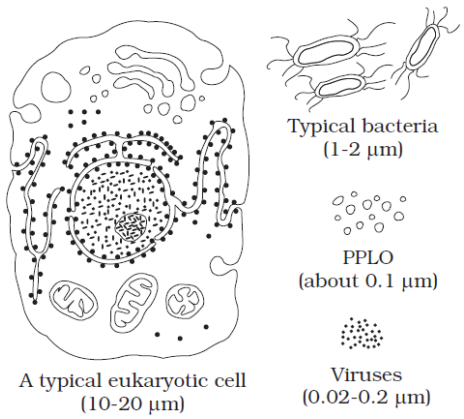


Diagram showing comparison of eukaryotic cell with other organisms

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123. (2)

Onion cell (a typical plant cell) and fungi possess cell wall; while white blood cells and mycoplasma lack them.

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124. (1)

Based on the position of the centromere, the chromosomes can be classified into four types. The metacentric chromosome has middle centromere forming two equal arms of the chromosome. The sub-metacentric chromosome has centromere slightly away from the middle of the chromosome resulting into one shorter arm and one longer arm. In case of acrocentric chromosome the centromere is situated close to its end forming one extremely short and one very long arm, whereas the telocentric chromosome has a terminal centromere.

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125. (2)

The material of the nucleus stained by the basic dyes was given the name chromatin by Flemming. Chromatin contains DNA and some basic proteins called histones, some non-histone proteins and also RNA.

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126. (3)

A single human cell has approximately two metre long thread of DNA distributed among its forty six (twenty three pairs) chromosomes.

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127. (4)

Most prokaryotic cells, particularly the bacterial cells, have a chemically complex cell envelope. The cell envelope consists of a tightly bound three layered structure i.e., the outermost glycocalyx followed by the cell wall and then the plasma membrane.

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128. (3)

Bacteria can be classified into two groups on the basis of the differences in the cell envelopes and the manner in which they respond to the staining procedure developed by Gram viz., those that take up the gram stain are Gram positive and the others that do not are called Gram negative bacteria.

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129. (2)

The genetic material in prokaryotes is basically naked, not enveloped by a nuclear membrane. In addition to the genomic DNA (the single chromosome/circular DNA), many bacteria have small circular DNA outside the genomic DNA. These smaller DNA are called plasmids. The plasmid DNA is used to monitor bacterial transformation with foreign DNA. The plasma membrane is selectively permeable in nature and interacts with the outside world. This membrane is similar structurally to that of the eukaryotes.

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130. (4)

The cell wall of a young plant cell, the primary wall is capable of growth, which gradually diminishes as the cell matures and the secondary wall is formed on the inner (towards membrane) side of the cell. The middle lamella is a layer mainly of calcium pectate which holds or glues the different neighbouring cells together. The cell wall and middle lamellae may be traversed by plasmodesmata which connect the cytoplasm of neighbouring cells.

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131. (2)

A number of organised flattened membranous sacs called the thylakoids, are present in the stroma of chloroplast. In both prokaryotic and eukaryotic cells, a semi-fluid matrix called cytoplasm occupies the volume of the cell. The inner compartment of mitochondria is filled with a dense homogeneous substance called the matrix. The interphase nucleus (nucleus of a cell when it is not dividing) has highly extended and elaborate nucleoprotein fibres called chromatin, nuclear matrix (nucleoplasm) and one or more spherical bodies called nucleoli.

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132. (2)

Microvilli are finger-like projections in the animal cells.

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133. (3)

Reserve material in prokaryotic cells are stored in the cytoplasm in the form of inclusion bodies. These are not bound by any membrane system and lie free in the cytoplasm e.g., phosphate granules, cyanophycean granules and glycogen granules. Gas vacuoles are found in blue green and purple and green photosynthetic bacteria.

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134. (3)

The Golgi cisternae are concentrically arranged near the nucleus with distinct convex *cis* or the forming face and concave *trans* or the maturing face. The cis and the *trans* faces of the organelle are entirely different, but interconnected. Materials to be packaged in the form of vesicles from the ER fuse with the *cis* face of the golgi apparatus and move towards the maturing face.

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135. (2)

An elaborate network of filamentous proteinaceous structures consisting of microtubules, microfilaments and intermediate filaments present in the cytoplasm is collectively referred to as the cytoskeleton. The cytoskeleton in a cell are involved in many functions such as mechanical support, motility, maintenance of the shape of the cell.

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136. (2)

Some of the columnar or cuboidal cells get specialised for secretion and are called glandular epithelium. They are mainly of two types: unicellular, consisting of isolated glandular cells (goblet cells of the alimentary canal), and multicellular, consisting of cluster of cells (salivary gland).

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137. (4)

Cockroach has compound eyes with many ommatidia; the resulting mosaic vision is highly sensitive and suited to nocturnal activity (low resolution, high sensitivity).

The supra-oesophageal ganglion (brain) innervates the antennae and compound eyes.

Besides Malpighian tubules, the fat body, nephrocytes, and uricose glands contribute to excretion.

Collateral glands are female accessory glands that secrete materials for the ootheca. Males have other accessory glands (e.g., mushroom/utricular and conglobate), not collateral glands.

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138. (2)

List-I	List-II
Chitinous exoskeleton	Provides mechanical support and protection
Thorax	Carries two pairs of wings
Abdomen	Segmental arrangement in both sexes, shows 10 divisions.
Jointed legs	Enables locomotion with flexibility

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139. (3)

Frog blood contains RBCs (nucleated), WBCs, and platelets.

Lymph lacks RBCs and has lower protein content than blood plasma.

Kidneys are compact, dark red and bean-like structures situated a little posteriorly in the body cavity on both sides of vertebral column.

In male frogs, the ureters carry both urine and sperm and open into the cloaca (urinogenital ducts).

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140. (1)

Male frogs have vocal sacs (for calling during breeding) and a nuptial/copulatory pad on the first digit of the forelimb to grip the female in amplexus—these are sexually dimorphic traits.

Both sexes adults lack neck and tail, have five-toed hind limbs, and possess bulging eyes with a nictitating membrane.

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141. (2)

Cartilage is avascular—it has no blood vessels within its matrix. Nutrients and oxygen reach chondrocytes only by diffusion from the perichondrium/synovial fluid, which is slow and inefficient. As a result, cell turnover and matrix repair are sluggish, so healing after injury is poor.

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142. (3)

Adult frogs breathe on land via pulmonary respiration (lungs), bucco-pharyngeal respiration (diffusion across the moist buccal mucosa with buccal pumping), and cutaneous respiration (across the moist, highly vascular skin).

During hibernation (winter) and aestivation (summer), frogs are largely inactive; cutaneous gas exchange sustains them as skin remains moist and well vascularized, while lung ventilation is minimal.

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143. (2)

Stratified epithelium is primarily protective; secretion/absorption roles are limited.

A diffusion boundary is chiefly the job of simple squamous epithelium (e.g., alveoli, endothelium), not stratified.

Blood vessels are lined by endothelium, which is simple squamous, not stratified.

In stratified epithelium, only the basal layer touches the basement membrane; upper layers rest on cells below.

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144. (3)

Plasma proteins are chiefly albumin, globulins, and fibrinogen—functional proteins for transport, immunity, and clotting. "Structural proteins" (e.g., collagen, elastin) are not plasma constituents.

In typical connective tissues, cells (e.g., fibroblasts) secrete modified polysaccharides /GAGs for the ground substance. Blood cells do not secrete the plasma matrix; plasma is largely water with proteins, electrolytes, nutrients, etc.

The fluid mediating exchange between tissues and blood is interstitial fluid/lymph. Blood is the main circulating fluid; it is not called lymph.

RBCs and platelets constitute maximum part of blood and are enucleated.

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145. (4)

- The organism famously known as the "friend of farmers" is the earthworm, due to its role in vermicomposting, improving soil fertility, and aeration. Cockroaches, on the other hand, are universally regarded as pests. They do not control other pests; they are the pests themselves.
- They are omnivorous and will eat a wide variety of materials, including stored food products, which they both consume and contaminate with their excretions. They are known mechanical vectors, capable of carrying and transmitting various bacterial and viral disease.

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146. (4)

Simple epithelium is one cell thick, resting on a basement membrane, with cells closely apposed via junctions (tight, adherens, desmosomes) and a thin intercellular cement—forming a continuous single layer for diffusion, filtration, secretion, or absorption. Simple epithelium have high regenerative capacity.

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147. (2)

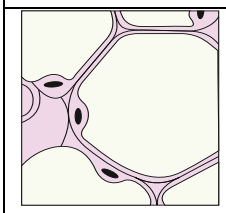
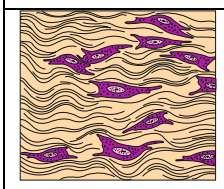
Frogs respire on land and in the water by two different methods. In water, skin acts as aquatic respiratory organ (cutaneous respiration). On land, the buccal cavity, skin and lungs act as the respiratory organs.

Frog heart has three chambers, two atria and one ventricle and is covered by a membrane called pericardium. A triangular structure called sinus venous joins the right atrium.

Frogs are generally ureotelic (excreting nitrogenous waste as urea) in their adult life on land. They do not change to uricotelic, which is a characteristic of animals that need to conserve water, like birds and reptiles. It's the tadpole, living in water, that is ammonotelic (excreting ammonia), and it turns ureotelic as it metamorphoses into a terrestrial adult.

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148. (4)

List-I	List-II
	Triglyceride storage
	Connect skin to muscle
	Present in skin
	Arrangement of fibres in ligament

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149. (3)

- Ciliated epithelial cells possess fine hair-like projections called cilia on their free surface. These cilia beat in a coordinated manner to move substances (like mucus, dust particles, or gametes) in a specific direction.
- In Bronchioles, Cilia move mucus, trapped dust, and microbes upwards toward the pharynx which keeps the airway clear and prevents infections.
- In Fallopian tubes, cilia help in the movement of ova from the ovary towards the uterus which is crucial in fertilisation and early zygote transport.
- The pancreatic duct is not lined with ciliated epithelium. Smaller pancreatic duct is lined with simple columnar or cuboidal epithelium, which is primarily involved in secretion. Larger pancreatic duct is lined with compound epithelium.
- The Eustachian tube is lined with ciliated epithelium to drain fluid from the middle ear. Salivary ducts are lined with stratified cuboidal epithelium and lack cilia.
- The bile duct is lined with simple columnar epithelium and does not have cilia. Its function is to transport bile, not to move mucus or particles with cilia.

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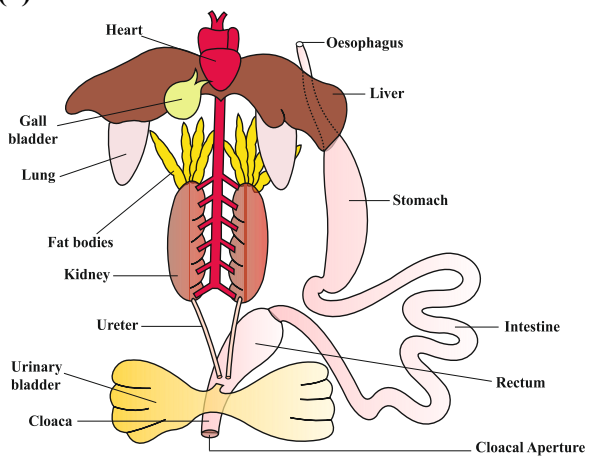
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150. (1)

- The first and second pair of thoracic spiracles and eight pairs of abdominal spiracles are regulated by muscles and can be opened and closed. The spiracles are not permanently open.
- Contraction of the tergo-sternal muscles compresses the abdomen, forcing air out of the respiratory system, a process known as expiration.
- The walls of the tracheal tubes are reinforced by rings of chitin, a tough polysaccharide, which prevents the tubes from collapsing.
- Cockroaches have an open circulatory system, and the tracheal system delivers oxygen directly to the tissues. Therefore, a respiratory pigment-like hemoglobin is not needed to transport oxygen in the hemolymph (blood).

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151. (4)



Diagrammatic representation of internal organs of frog showing complete digestive system

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152. (1)

In pseudostratified epithelium, all cells touch the basement membrane, but nuclei lie at different heights, creating a false stratified look.

It is actually a single layer (simple) epithelium, not multiple layers, and its main roles (e.g., in the respiratory tract) are secretion and mucociliary clearance, not heavy protection from stress (that's a role of stratified epithelia).

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153. (3)

- Cockroaches are invertebrates belonging to the phylum Arthropoda, the largest phylum in the animal kingdom. They are further classified under the class Insecta, which includes all insects.
- The body of a cockroach is distinctly divided into three main regions: the head, thorax, and abdomen.
- The head of *Periplaneta americana* (the American cockroach) is triangular in shape, but it is bent at a right angle (90 degrees) to the longitudinal body axis, not an acute angle. This orientation allows it to look straight down as it feeds.
- In frogs, bile is secreted by the liver and stored in the gallbladder. Its primary function is to emulsify fats, breaking them down into smaller droplets. The pancreatic juices, secreted by the pancreas, contain enzymes that digest both carbohydrates and proteins.

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154. (2)

- An organ system represents a higher level of organization in complex multicellular organisms. It is formed when several organs, each with a specialized function, work together to achieve a broader physiological purpose. This cooperation is a key aspect of the division of labor in complex animals.
- For example, the digestive system is an organ system that includes the stomach, small intestine, liver, and pancreas. The stomach breaks down food, the small intestine absorbs nutrients, and the liver and pancreas secrete enzymes and bile to aid digestion. All these organs work together to process food and extract nutrients for the body.
- A single cell or tissue performing all functions, is characteristic of simple organisms, not complex ones with organ systems.
- A tissue is a group of similar cells working together, whereas an organ system is a group of organs working together.

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155. (2)

- The primary function of connective tissue is to bind and support other body tissues and organs. This is the origin of their name.
- The examples—cartilage, bone, adipose (fat), and blood—are all specialized types of connective tissue. This diversity reflects their wide range of functions.
- Strength, elasticity, and flexibility are provided by the extracellular matrix, which is a key component of connective tissue. The matrix contains a variety of protein fibers and ground substance.
- The cells of connective tissue, such as fibroblasts and osteocytes, secrete structural protein fibers like collagen and elastin. Collagen provides tensile strength, while elastin provides elasticity.

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156. (4)

- Neural tissue, composed of neurons and neuroglia, is specialized for control and coordination. Neurons are excitable cells that detect stimuli, transmit information, and generate responses. This system integrates the functions of all other organ systems.
- The inability of the coma patient to respond demonstrates that even if all other tissues (muscular, connective, epithelial) are technically functional, a coordinated, meaningful response to the environment is impossible without a properly functioning neural tissue. The neural system is the master controller that integrates sensory input with motor output. Its failure leads to a loss of responsiveness, proving its central role.

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157. (4)

- The eyes and internal ears are indeed well-organized, the remaining sensory structures for touch and taste are cellular aggregates around nerve endings.
- The division of labor among cells, tissues, and organs is a characteristic of complex multicellular organisms. This specialization actually increases efficiency and allows for faster and more coordinated survival responses, rather than slowing them down. It also reduces metabolic redundancy because individual cells do not need to perform all life functions.

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158. (4)

List-I	List-II
Gall bladder in frog	Temporary storage reservoir for bile produced by the liver
Dendrites	Nerve cells
Bundle of muscle fibres	Enclosed in a sheath of connective tissue
Haversian canal	Mammalian bone

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159. (1)

- All glands in the body (lacrimal for tears, mammary for milk, sudoriferous for sweat, sebaceous for oil) are derived from and are a type of glandular epithelium. Therefore, secretion is a major function of epithelial tissue.
- The exposed surface of the body is the skin. The outermost layer of the skin, the epidermis, is a classic example of stratified (compound) squamous epithelium (specifically, keratinized stratified squamous epithelium).
- A cockroach has one pair of salivary glands, not two pairs.
- The name "reservoir" in salivary reservoir accurately describes its function. It is a bladder-like structure that stores saliva produced by the glands until it is needed for feeding.
- The description of the antenna in the question is correct, but the technical term is wrong. "Thread-like" antennae are called filiform. "Moniliform" antennae are bead-like, as seen in termites.

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160. (4)

List-I	List-II
Prevents leakage of body fluids between adjacent cells	Tight junctions
Provides cytoplasmic continuity between neighbouring cells	Gap junctions
Acts as anchoring points that bind cells together	Adhering junctions
Connects one bone to another	Ligaments

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161. (4)

- Transitional epithelium is a compound epithelium that consists of multiple layers. It is present in urinary bladder. It has property of being able to stretch.
- The vibrant coloration of certain tropical cockroach species is generally not an example of sexual dimorphism, but rather serves as a warning signal to predators, a defense mechanism known as aposematism. These colors are often found in both males and females.

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162. (3)

- A hole in the floor of the buccal cavity makes it impossible for the frog to create a sealed chamber. This disrupts the buccal pump mechanism, which is essential for two modes of respiration:
  - Pulmonary Respiration: The frog cannot build up the positive pressure required to force air from its mouth into its lungs.
  - Buccal Respiration: The frog cannot hold air in its mouth to allow for gas exchange across the moist lining of the buccal cavity.
  - Therefore, both pulmonary and buccal respiration will effectively cease.
- The frog has a third mode of respiration: cutaneous respiration. Its moist, highly vascular skin allows for the direct diffusion of gases between the blood and the environment (air or water). This process is completely independent of the mouth and lungs. In a resting state, cutaneous respiration can be sufficient to meet the frog's metabolic oxygen demands, allowing it to survive even when its other respiratory mechanisms are non-functional.

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163. (4)

List-I	List-II
Mast cells	Areolar connective tissue
Hyaline cartilage	Tip of nose
Crop	Temporary storage of food
Tubular parts of nephron	Cuboidal epithelium

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164. (2)

- A (Labrum) is the upper lip → only holds food, does not chew.
- B (Left mandible) and C (Right mandible) are the paired cutting and grinding structures in cockroach. Their toothed edges help in chewing and crushing food.
- D (Hypopharynx) acts as a tongue → directs food.
- E (Labium) is the lower lip → helps in holding food, not grinding.
- F and G (Maxillae) → aid in handling and tasting food.

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165. (3)

- Skeletal muscle tissue is composed of numerous long, cylindrical fibers, known as muscle cells, which are organized in parallel bundles. This arrangement is crucial for generating force and producing coordinated movement.
- The fine fibrils within muscle fibers are called myofibrils, not neurofibrils. Neurofibrils are found in neurons (nerve cells).
- After contracting and shortening, muscle fibers relax and return to their original relaxed (lengthened) state, not a "contracted state."
- Muscles are responsible for all forms of movement, whether it's voluntary movement to respond to the environment or involuntary actions to maintain posture and position.
- Muscles are the primary movers of the body and play an active role in all forms of movement, from walking and running to the beating of the heart and the movement of food through the digestive tract.

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166. (3)

List-I	List-II
Mouth parts of Cockroach	Mandibles, maxillae
Segments of walking leg	Coxa, trochanter
Sound receptor structure	Anal cerci
Excretory unit	Malpighian tubule

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167. (3)

- The nervous system of a cockroach consists of a series of fused, segmentally arranged ganglia joined by paired longitudinal connectives on the ventral side of the body.
- Maxillary and labial palps in cockroaches are sensory structures that help in tasting and touching.
- The supra-oesophageal ganglion, a mass of nervous tissue located above the esophagus, serves as the brain in cockroaches, controlling most of their sensory and motor functions.
- Mosaic vision, a type of vision where many individual lenses (ommatidia) create a single image, has high sensitivity (especially in low light) but less resolution.

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168. (3)

- The pronotum is the large, shield-like dorsal plate of the prothorax, the first segment of a cockroach's thorax.
- The oesophagus is a narrow food tube that connects the pharynx to the crop, a storage sac for food.
- An ocellus is a simple eye that can only detect changes in light intensity and direction, not form a sharp, high-resolution image. That function is carried out by the more complex compound eyes.
- The gizzard, or proventriculus, in a cockroach has six chitinous plates that act like teeth. These plates grind the food before it enters the midgut.

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169. (3)

The stomach has an epithelial lining for secretion and absorption, muscle for churning food, connective tissue for support, and nerves to control these functions.

The heart is predominantly muscle, but it also has an epithelial lining (endothelium), connective tissue in its valves and outer layers, and a nervous tissue for regulating its beat.

The lungs are not just two tissue types; they are complex organs with:

- Epithelial tissue lining the air sacs (alveoli) for gas exchange.
- Connective tissue providing a framework.
- Smooth muscle in the bronchioles to regulate airflow.
- Nervous tissue to control breathing.

Similarly, the kidneys contain all four tissue types for their filtration function.

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170. (3)

- *Periplaneta americana*, the American cockroach, is a cosmopolitan pest and is very common in India.
- A cockroach does not breathe through its head. Respiration occurs through tiny holes called spiracles located along its body segments. Therefore, decapitation does not immediately stop breathing.
- A cockroach's nervous system is decentralized. It consists of a series of ganglia (nerve clusters) located in each body segment, with the supra-oesophageal ganglion in the head serving as a brain but not holding the entire nervous system.
- The entire body is covered by a hard chitinous exoskeleton (brown in colour). In each segment, exoskeleton has hardened plates called sclerites (tergites dorsally and sternites ventrally) that are joined to each other by a thin and flexible articular membrane (arthrodial membrane).

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171. (3)

- Adipocytes, or fat cells, in white adipose tissue are specialized for storing a large lipid droplet. This droplet pushes the cytoplasm and nucleus to the periphery of the cell, giving it a characteristic signet ring shape.
- The circulatory system of a cockroach is an open type. The dorsal heart is a long, tubular structure that extends along the thorax and abdomen. It is indeed composed of 13 funnel-shaped chambers, and its contractions are aided by 12 pairs of alary muscles that draw hemolymph into the heart. The pulsating heart and the alary muscles work together to circulate the hemolymph throughout the body cavity.

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172. (3)

- Frogs do not have salivary glands with a digestive function. Their saliva is primarily used to lubricate food for swallowing. Chemical digestion of carbohydrates begins in the small intestine, not the buccal cavity.
- The pacemaker of the frog's heart, like that of mammals, is composed of specialized cardiac muscle tissue (myogenic tissue) located within the heart wall itself. The neural input from the central nervous system only modifies the heart rate, it does not originate the heartbeat.
- The absence of a neck in a frog contributes to its streamlined body and compactness, which is an adaptation for its jumping lifestyle, but it is not the only contributor. The main adaptations for jumping are the powerful hind limbs and the rigid vertebral column.
- A frog's primary mode of water intake is not by drinking through the mouth. Instead, it absorbs water directly through its highly permeable skin, especially through a patch of skin on its belly known as the "drinking patch."

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173. (4)

- High cranial mobility (the ability to move the head) in a cockroach comes from the flexibility of its neck, not from its rigidity. The cockroach's head can move in various directions, which is essential for feeding and sensing its environment.
- The cockroach's head is indeed a product of the fusion of six embryonic segments, a process known as cephalization. This head then articulates with the thorax (specifically the prothorax) via a soft, membranous, and highly flexible neck, which allows for extensive movement.

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174. (4)

- **Collateral glands (A):** In females, they secrete the oothecal covering, which forms the protective case for fertilised eggs.
- **Spermatheca (B):** Functions as the sperm storage organ in the female reproductive tract, essential for fertilisation.

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175. (4)

When a cockroach is at rest, its delicate hind wings are folded like a fan and tucked beneath the protective forewings.

The forewings, called tegmina, are tough and leathery, acting as covers. The hind wings are thin, transparent, and used for flight.

The hind wings are the primary flight wings. They generate the lift and propulsion needed for flying.

The forewings are called tegmina, they arise from the mesothorax, not the metathorax. The hind wings arise from the metathorax.

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176. (3)

The male frog has a pair of testes attached to the upper part of the kidneys.

1. Sperm are produced in the testes.
2. They pass into a series of 10-12 fine tubules called the vasa efferentia.
3. Crucially, these vasa efferentia enter the kidney on the same side.
4. Inside the kidney, they open into a tube called the Bidder's canal.
5. From the Bidder's canal, the sperm travel through the collecting tubules and are finally carried out of the kidney by the ureter.
6. In male frogs, the ureter acts as a common urinogenital duct, carrying both urine from the kidney and sperm from the testes.
7. The ureter opens into the cloaca, from where the sperm are expelled.

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177. (2)

List-I	List-II
Moist surface of pharynx	Stratified non keratinised squamous epithelium
Larger ducts of salivary glands	Stratified cuboidal epithelium
Ventricles of Brain	Ciliated columnar epithelium
Lining of stomach	Non ciliated columnar epithelium

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178. (1)

- In the male reproductive system of a cockroach, the utricular glands consist of long and short tubules. These glands secrete substances that help nourish the sperm and contribute to the formation of the spermatophore, a capsule that contains and protects the sperm.
- The medulla oblongata is a part of the brain in vertebrates (like humans and frogs). The foramen magnum is a large opening at the base of the skull in vertebrates through which the spinal cord passes. These two terms are not directly related as a structure-function pair in a cockroach, which has a decentralized nervous system and no skull or spinal cord.
- Skin of frog on dorsal side is olive green with dark irregular spots.
- Cockroaches have three pairs of walking legs, not six pairs. Each pair arises from one of the three thoracic segments (prothorax, mesothorax, and metathorax).

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179. (3)

The nerve impulse arrives at the neuron's endings, which is the output zone (axon terminals), not the "input zone." The input zone consists of the dendrites and cell body, where a neuron receives signals.

Neurons are indeed excitable cells. However, they are not contractible. The property of contractility (the ability to shorten forcefully) is the defining characteristic of muscle cells, not neurons.

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180. (1)

- Mammalian blood has a much higher oxygen-carrying capacity than frog blood. A key reason for this is the difference in the structure of their red blood cells.
- Mature mammalian red blood cells (erythrocytes) are biconcave and enucleated (lacking a nucleus). The absence of a nucleus allows the cell to be packed with a higher concentration of hemoglobin, the protein responsible for binding oxygen. In contrast, a frog's erythrocytes are nucleated, and the nucleus takes up a significant amount of space, limiting the amount of hemoglobin they can hold. This structural difference directly results in the higher oxygen-carrying capacity of mammalian blood, which is essential for their high metabolic rate.

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